

spinmate

User Guide

An AI agent for spintronics simulation. Describe a setup in plain language or upload a parameter file; SpinMate validates it, runs the two-sublattice macrospin model and explains the result from the actual magnetization dynamics.

Version 1.0 · Free and open source (MIT) · github.com/Michaellianeris/sublattice-agent

Contents

1. Getting started
2. Asking for a simulation
3. Reading a run card
4. Fast and Deep
5. Models and the guide
6. Sweeps, threshold searches and continuing runs
7. Comparing runs, figures and reports
8. The Runs panel and questions across runs
9. Settings, cost and conversations
10. Parameter reference
11. Troubleshooting

1. Getting started

1. **Start the app.** With Docker: `docker compose up -d`, then open `http://localhost:8421`. On Windows you can also run `SpinMate.exe`, which opens the app in your browser.
2. **Meet Athena.** The first time, Athena (the owl) asks for your name and email. The email creates your account and brings you this guide; you can skip it.
3. **Add an API key.** Open Settings (gear icon) and paste an Anthropic and/or OpenAI key. A key in the page stays in that browser tab; a key in `.env` is used by the server for everyone on it.
4. **Ask.** Type a request in the message box and press Enter, or pick one of the cards under *Find a skill*.

> The simulator also works without a key: the play icon in the left bar (*Run setup*) runs a preset or your own parameter file directly. Only the chat needs a model.

2. Asking for a simulation

Describe the starting state, the exchange, any field, current or temperature, and the duration. Units follow the solver (SI; currents in A/cm²). Examples:

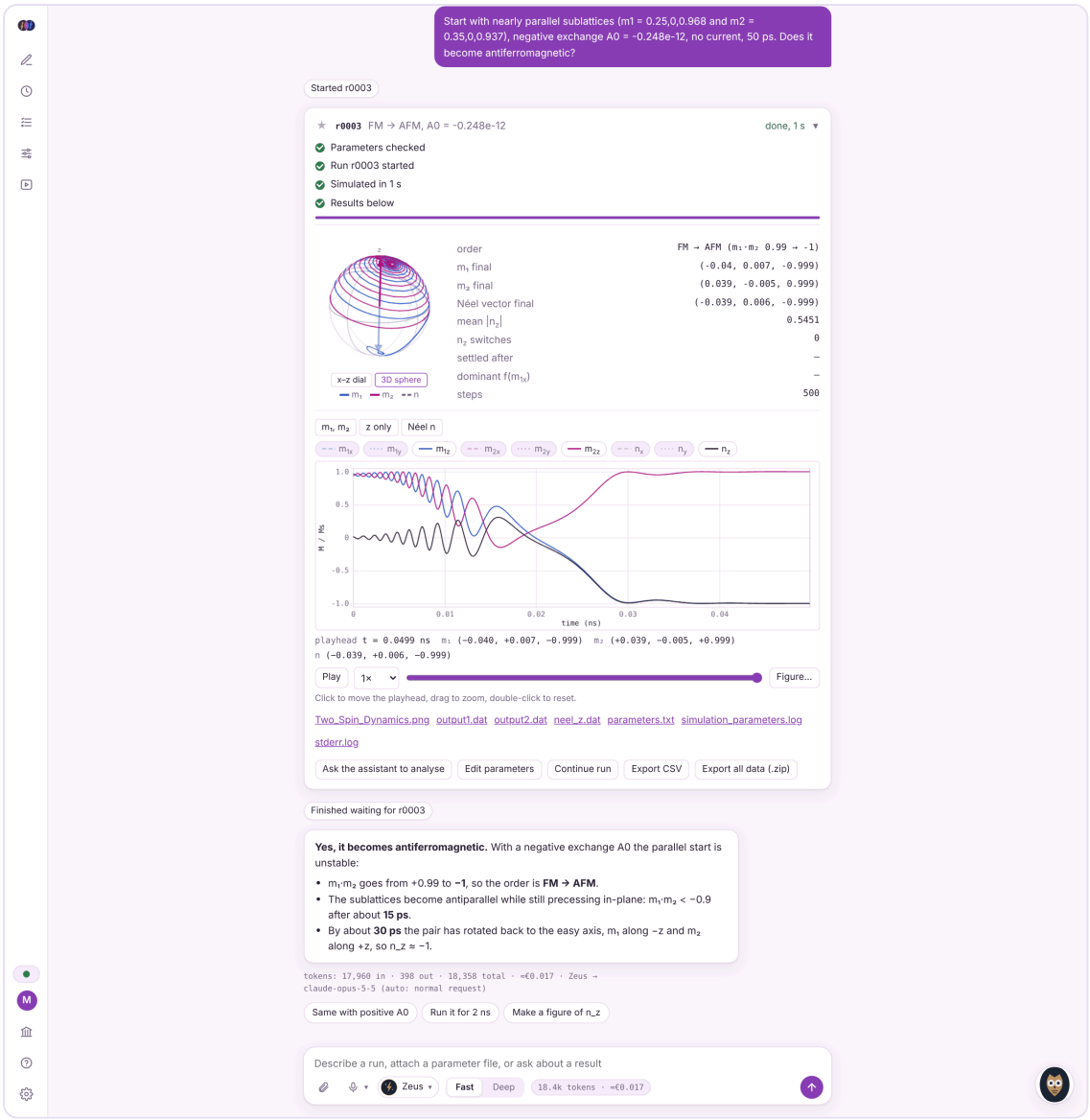
> *Start antiferromagnetic with a positive exchange A0 = 0.248e-12, no current, 50 ps. Does it end ferromagnetic?*

> *Start with nearly parallel sublattices (m1 = 0.25,0,0.968 and m2 = 0.35,0,0.937) and A0 = -0.248e-12, 50 ps.*

> *Apply a DC spin-orbit current of 100e6 A/cm² for 2 ns to the default AFM state. What happens?*

> *Heat a FeRh layer through its transition: positive A0, 370 K ± 60 K at 0.25 GHz, 2 ns.*

SpinMate checks the parameters with the solver's own parser before anything runs. A suspicious value (for example `A0 = 0.248` without the `e-12`) is pointed out instead of being run. Large requests (many runs or long runtimes) wait for your **Run it** button.



A request in plain language, the run card and the answer with suggested next steps.

3. Reading a run card

Field	Meaning
order	Initial and final order from $m_1 \cdot m_2$: AFM below −0.9, FM above +0.9.
m_1, m_2 final, Néel vector	Final unit vectors and $n = (m_1 - m_2)/2$.
n_z switches	Real reversals of n_z (from below −0.5 to above +0.5 or back), with the time of the first one. Ringing around zero is not counted.
settled after	When m_1 and m_2 stop changing (within 2 %). "–" if they did not settle in the run.
dominant f	Main frequency of m_{1x}, given only if at least five periods fit in the second half of the run.

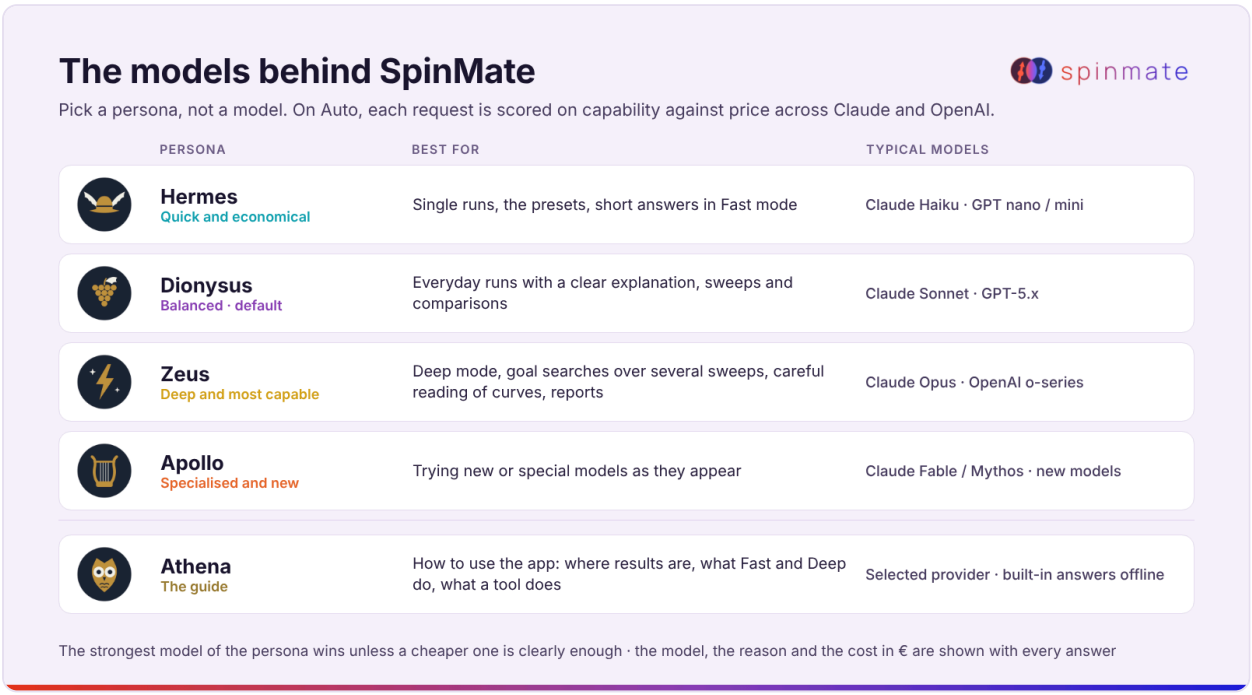
The **sphere** shows the trajectories of m_1 (blue) and m_2 (magenta) in 3D, or as an x–z dial. The **chart** shows m_1, m_2, z-components only, or the Néel vector; drag to zoom, double-click to reset, click to move the playhead. **Play** animates the run. The card also links the raw files, CSV export and a zip of all data.

4. Fast and Deep

Fast uses the default parameters, changes only what you name and answers in a few lines. **Deep** works from the full parameter set or your file, mentions the caveats that apply and gives a quantitative explanation. Both end with up to three suggested next steps as buttons.

5. Models and the guide

Under the message box you pick a persona, not a model. On *Auto* each request is scored on capability against price across your Claude and OpenAI keys; the strongest model of the persona wins unless a cheaper one is clearly enough. Every answer shows the model, the reason and the cost in €. To always use one model, pick it in Settings → *Model behind each god*.



6. Sweeps, threshold searches and continuing runs

- **Sweep:** “Sweep A0 from −0.25e-12 to +0.25e-12 in 11 steps, 0.5 ns each.” Runs go in parallel; the card shows a table and an overlay.
- **Threshold search:** “Find the smallest SOT current between 50e6 and 150e6 A/cm² that makes the Néel vector rotate.” SpinMate brackets the threshold with a few rounds of sweeps and reports the interval.
- **Continue a run:** “Continue r0003 for another 0.5 ns.” Time does not restart; *combine* joins the segments into one curve.

7. Comparing runs, figures and reports

- **Compare:** “Compare r0003 and r0005.” Lists the parameters that differ and overlays the curves.
- **Figures:** “Make an IEEE figure of n_z for r0001 and r0003, first 60 ps.” Styles: science, IEEE, Nature. The figure appears in the chat with PDF, SVG and PNG downloads, and *Change…* to adjust it.
- **Reports:** “Write a short report of these runs with your interpretation.” The report has the SpinMate header, method, results table, parameters, figures and appendix, with LaTeX math. Print it to PDF or download the Markdown version.

8. The Runs panel and questions across runs

The list icon in the left bar opens every run: search by id, label, #tag or parameter, filter by status and end state, star and tag runs, select several to compare, plot or report, or delete them. With the run catalog enabled, the assistant answers questions across runs with one read-only SQL query:

> *Make me a table of all finished runs: id, A0, initial and final order, first n_z switch in ps.*

9. Settings, cost and conversations

- **Theme:** Spin (default), Light or Dark in Settings.
- **Default parameters:** saved per user and used as the base of every run.
- **Cost:** the counter next to the mode buttons shows tokens and the estimated cost in €, converted with the daily ECB rate.
- **Conversations:** the clock icon lists your chats; select several or all and delete them. Runs are kept.
- **Voice:** the microphone records a request; the arrow next to it chooses the microphone and language.

10. Parameter reference (most used)

Parameter	Meaning	Example
`--t`	Duration (s)	`--t=5e-11`
`--A0`	Inter-sublattice exchange (J/m); negative = AFM coupling	`--A0=-0.248e-12`
`--m1`, `--m2`	Initial unit vectors	`--m1=0.25,0.0,0.968`
`--SOT_DC_Amp`	DC spin-orbit current (A/cm²)	`--SOT_DC_Amp=1e8`
`--flagTempVarying`, `--Temp`, `--Fr`	FeRh temperature cycle: on/off, amplitude (K), frequency (Hz)	`--Temp=60 --Fr=0.25e9`

A parameter file is plain text with one `--name=value` per line; attach it with the paper clip. Ask Athena or the assistant “Which parameters can I set?” for the full list with units.

```
--t=2e-9
--A0=-0.248e-12
--m1=0.25,0.0,0.968
--m2=0.35,0.0,0.937
```

11. Troubleshooting

Symptom	What to do
“Add an API key”	Paste a key in Settings, or set it in `.env` and restart.
No answer, 401/429 errors	Check the key and the credit of the provider; on Auto the next model by score is tried.
Page does not open from another computer	By design the app listens on this computer only (`BIND_ADDRESS=127.0.0.1`).
Behind a corporate proxy	Put the company CA bundle in `./certs` and set `SSL_CERT_FILE` (see README).
Microphone not listed	Allow the microphone in the browser once; open the app on localhost or https.

More in the README on GitHub. Feedback and issues are welcome.